

Stratum: illustrated beginner's guide

1. Get to know the workspace

Import a recording, create useful results, and keep every step traceable. Original samples stay unchanged; each operation creates new signals or values.

Start with the example

Launch Stratum and choose **Workspace > Open example workflow**, or choose **Open example workflow** in an empty workspace. The motor-test example loads seven completed steps, from original signals to smoothing, segments and values.

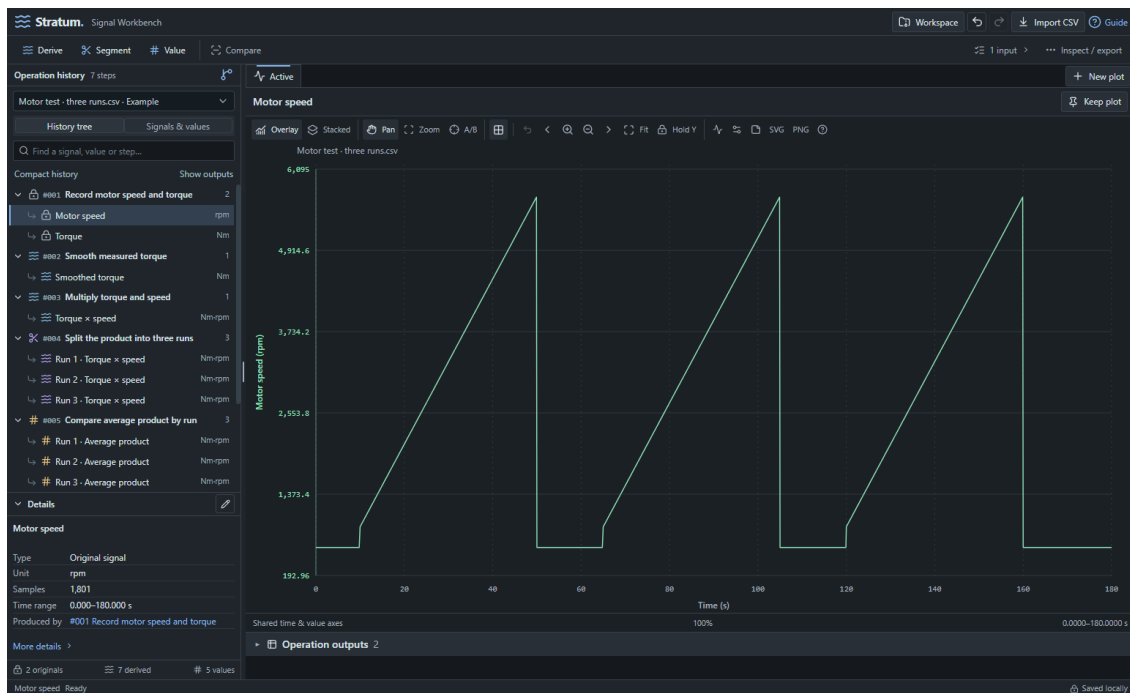


Figure 1. History is on the left; processing tools run across the top. Active shows the selected signal, and Operation outputs sits below the plot.

Click the arrow beside a step in **History tree** to expand its outputs, then click a signal name. **Signals & values** offers another way to find a result. The **Scope** selector above the history switches between recordings.

Import your own data

Choose **Import CSV**. Use a comma-separated UTF-8 file, with time in seconds in the first column and units in square brackets in the header:

```
Time [s],Speed [rpm],Torque [Nm]
0.00,1500,210
0.01,1502,211
0.02,1504,212
```

Times must increase strictly, with no duplicates. Leave missing signal cells empty. Work is saved locally on this device. The screenshots in this guide use the included example, so you can follow along without importing a file.

2. Choose inputs and smooth a signal

The next three pages walk through one small workflow: smooth torque, keep a 10-50 second interval, then calculate its average.

Select the original torque signal

Expand **#001 Record motor speed and torque** and click **Torque**. It appears in **Active**. Check the input-count control near **Inspect / export**: it should show **1 input** for this exercise.

Remember: viewing an item and checking processing inputs are separate actions. Checked inputs stay selected while you browse. Open the input-count control and choose **Follow selection** to use the item you are viewing again.

Apply a moving average

Choose **Derive > Filters > Moving average**. Set **Window size** to **5** samples, as shown below. Expand the **review selection** disclosure if you want to check the input name, then choose **Create 1 derived signal**.

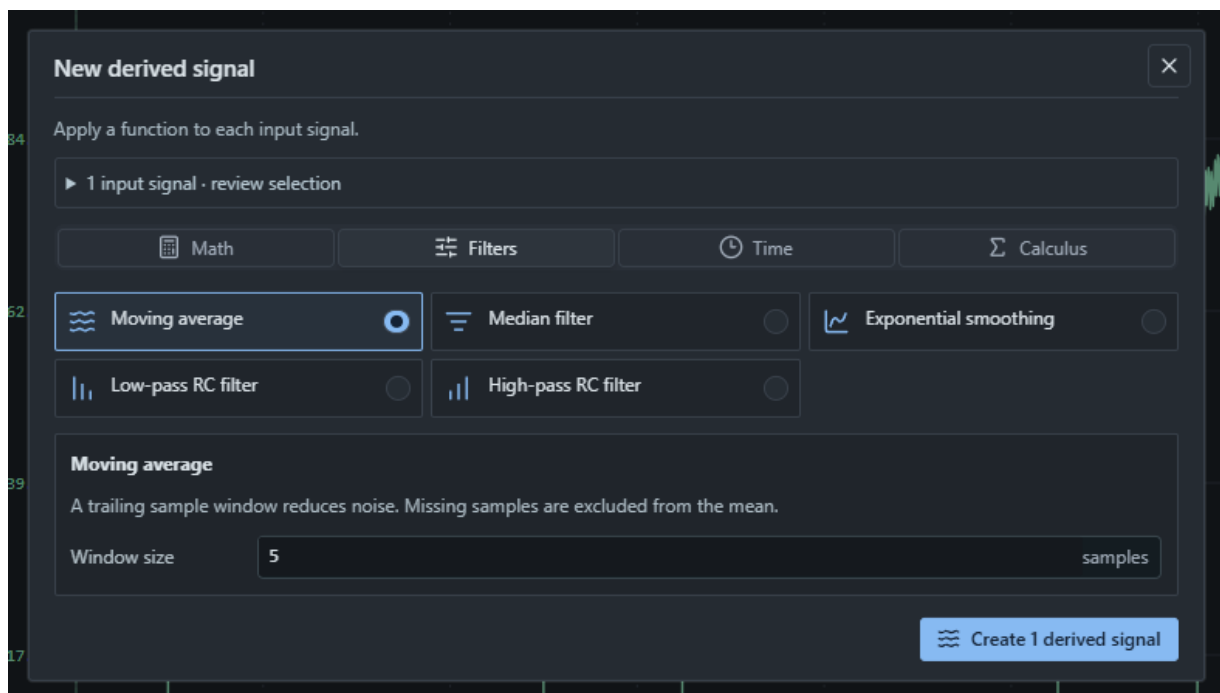


Figure 2. The Filters tab contains Moving average. Window size controls how many samples are used; the button creates a new derived signal.

The new operation appears at the end of history. Select its **Torque · Smoothed** output before continuing. The original Torque signal remains available under step #001.

Process several signals together

Select a step and expand **Operation outputs** below the Active plot to reveal the **Step outputs** table. Tick the signals you want to process, then check the input count before opening Derive, Segment or Value. Use **Follow selection** when you want to return to a single selected signal.

3. Keep a useful time interval

A segment is a derived signal covering part of its input. You can process it further or segment it again.

With the new smoothed output selected, choose **Segment > Time ranges**. Click **Add exact range**, set **Start (s)** to **10** and **End (s)** to **50**, then choose **Preview**.

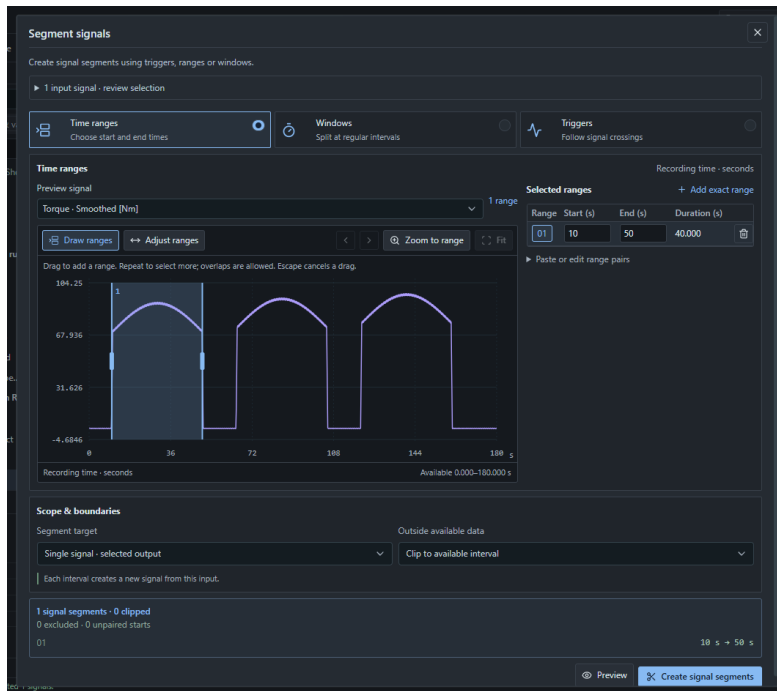


Figure 3. The shaded band and exact fields describe the same interval. The preview confirms one segment from 10 s to 50 s, with no clipped intervals.

Check **Segment target** and the preview boundaries, then choose **Create signal segments**. Scroll down in the dialog if the buttons are below the visible area. Select the new segment in history for the next step.

You can also drag across the plot with **Draw ranges**, or move and resize a range with **Adjust ranges**. **Windows** splits at regular intervals; **Triggers** uses signal threshold crossings. Start with Time ranges while learning the workflow.

4. Turn a signal into a value

Select the segment you just created. Choose **Value > Time average**, review the input, then choose **Create 1 value**.

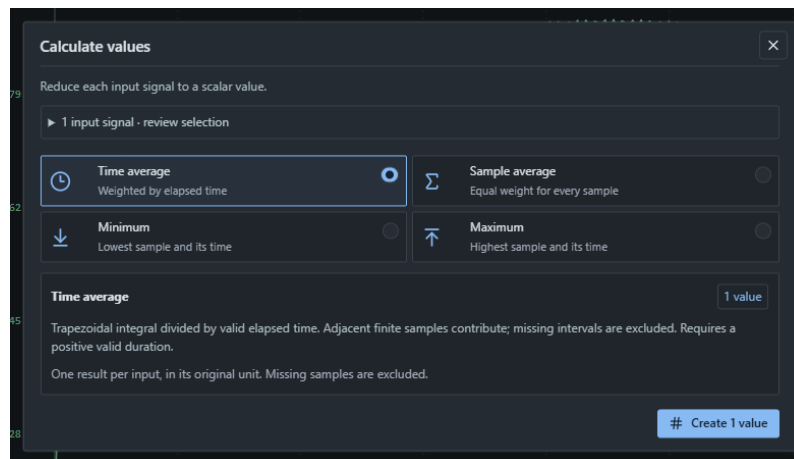


Figure 4. Value creates one number per input signal. The calculation cards explain how each result is calculated.

Open the new value in history. The result card shows its value, unit, finite sample count and valid time coverage; the plot below shows its input signal.

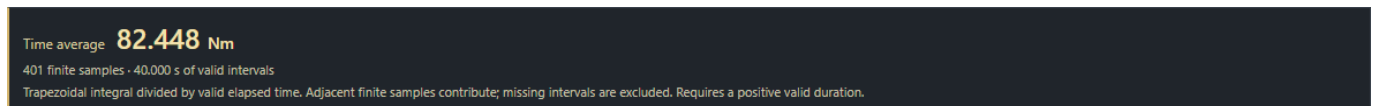


Figure 5. The result card from this example exercise. Your own recordings will produce different values and coverage.

Time average weights by elapsed time and excludes missing intervals. **Sample average** gives each finite sample equal weight. **Minimum** and **Maximum** find the lowest and highest finite samples.

Trace and revise your work

Open **Inspect / export** for **View samples**, **Inputs and originals**, **Show lineage in tree**, or **Used by later operations**. Clear search or choose **Show all steps** if a filter hides something you expect to see.

Right-click a History item to rename it or manage its operation. **Edit settings** updates the operation and recalculates dependent results; **Duplicate operation** starts a separate branch. **Delete operation** previews the dependent work it will also remove. The header's **Undo** and **Redo** keep the last 20 workspace changes, including across restarts.

5. Compare signals on a plot

Active follows the item you inspect. **Keep plot** creates a named plot that stays open while you browse, and **New plot** starts a blank comparison.

Compare the three example runs

Choose **New plot > Add signals**. Find the three **Run 1**, **Run 2** and **Run 3** signals named **Torque × speed**, tick them, then choose **Apply signals**. Turn on **Align starts at 0** to compare their elapsed time.

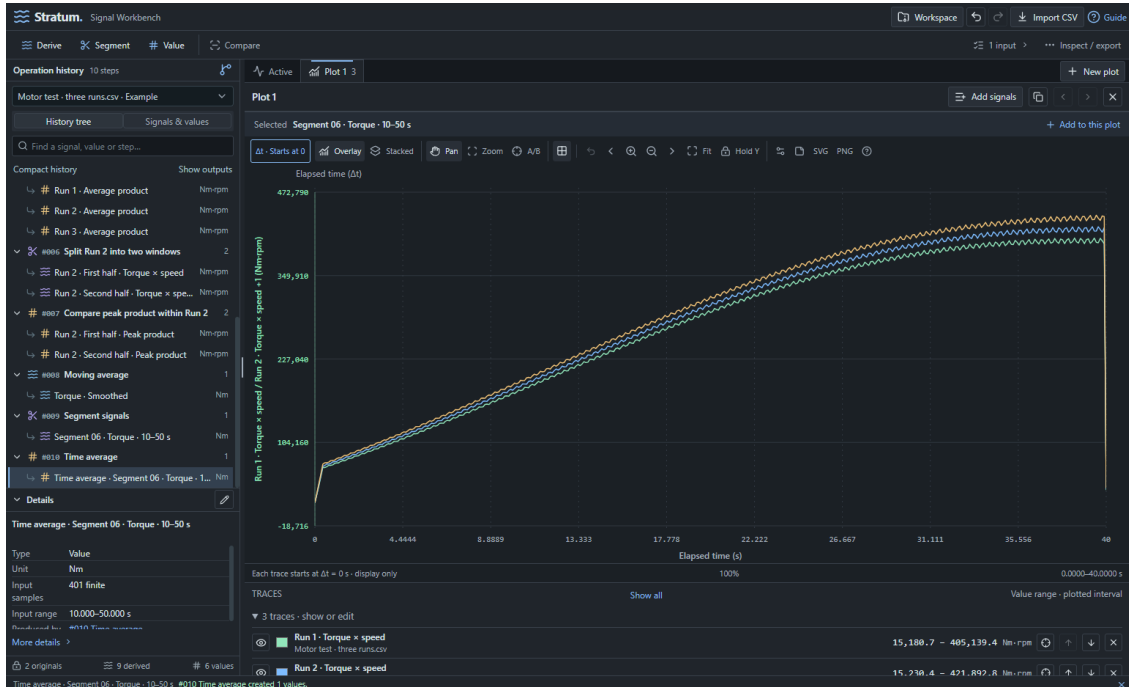


Figure 6. Three runs share an elapsed-time axis. The Starts at 0 control is active; the numbered badge on Plot 1 shows that it contains three traces.

Align starts at 0 changes the display only. **Compare** opens **Compare & align** when you want to create reusable aligned signals, including signals from different recordings.

Navigate and adjust the comparison

- Click the plot to focus it, then scroll to zoom time. Shift-drag pans in time. Choose **Fit** to see the complete plot again.
- Use **Overlay** for a shared plot or **Stacked** for separate panels. Open the trace list below the plot to show, hide or edit individual traces.
- Drag signals onto a plot to add them. Dragging a segment adds all sibling segments from its operation; **Add signals** lets you choose individual members.
- Use the plot toolbar's **SVG** or **PNG** controls to save an image of the plot.

Named plot layouts are saved on this device, separately from workflow history and workspace backups.

6. Export results and keep a backup

Choose the results to export

Choose **Inspect / export > Export / report**. Set **Include** to the viewed output, checked signals, or all outputs from a step. Choose a **File format**, review the included outputs, then choose **Download file**.

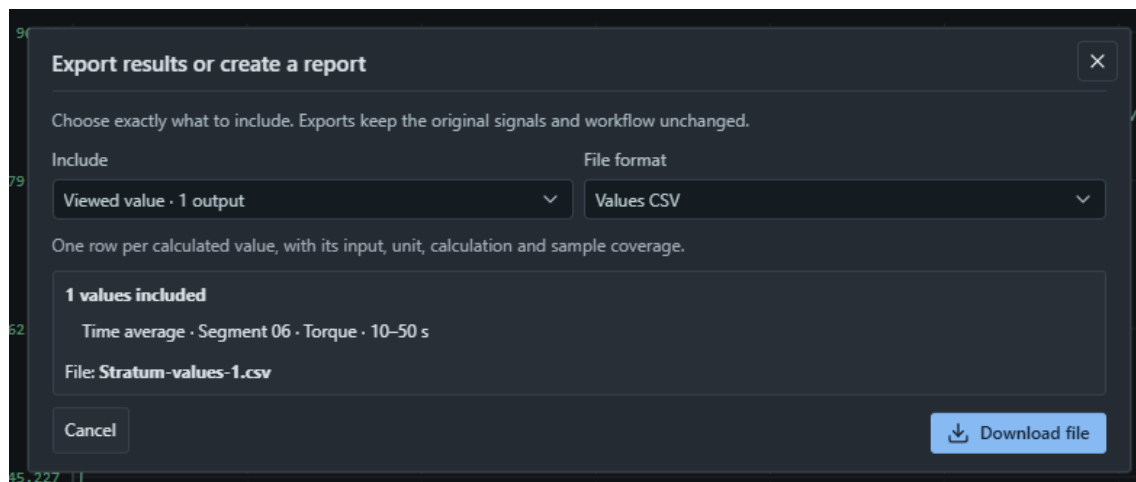


Figure 7. This export contains just the viewed value. Check Include before downloading so you get the intended output or batch.

Format	Use it for
Samples CSV	Every evaluated sample, with its time axis; missing values stay blank.
Values CSV	One row per calculated value, with units and its input.
Signal summary CSV	A compact summary of each signal, without individual samples.
Printable report (HTML)	A snapshot of results, plots and contributing history. Open in a browser to print or save as PDF.

Save the whole workspace

Choose **Workspace > Download workspace backup** and keep the **.stratum** file somewhere safe. Older **.stratus** backups can still be restored.

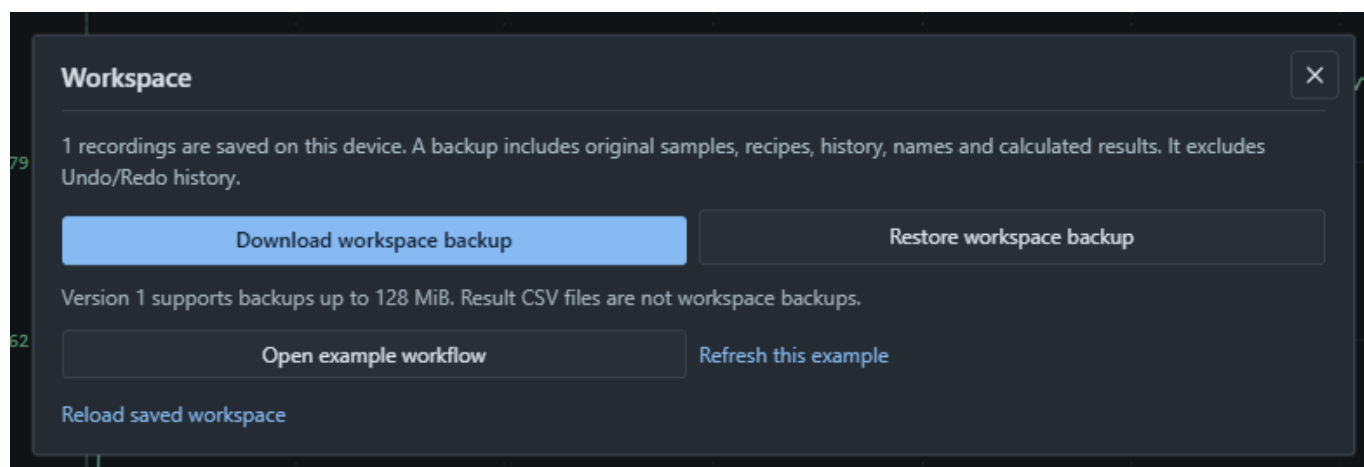


Figure 8. Workspace backups preserve the analysis for later restoration. Result CSV files are not workspace backups.

A backup includes original samples, recipes, results, names and history; saved plot layouts and Undo/Redo history are excluded. **Restore workspace backup** replaces the current workspace after confirmation; Undo can recover the prior one.

Desktop and browser workspaces have separate local storage. Use a backup to move your analysis between them or to another computer. Backups support up to **128 MiB**; Samples CSV supports **64 MiB** per file. For a larger sample export, choose fewer signals or shorter segments. The header's **Guide** opens the built-in workflow help.